

Signal Power

- A signal with *finite and non-zero power (mean square value)* is a **power signal**.

➤ Time average of the energy signal [for real valued signals],

$$P_g = \lim_{T_0 \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g^2(t) dt$$

➤ A signal is a power signal if,

$$0 < \lim_{T_0 \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |g(t)|^2 dt < \infty$$

- ❖ A necessary condition for the energy to be finite is that the amplitude $\rightarrow 0$ as $|t| \rightarrow \infty$, *if it does not*, then the signal energy is infinite and time average of the energy is more meaningful, called **average power**.

Signal Power and Power Spectral Density (PSD)

$$P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g^2(t) dt$$

The signal power and the related concepts can be readily understood by defining a truncated signal $g_T(t)$ as

$$g_T(t) = \begin{cases} g(t) & |t| \leq T/2 \\ 0 & |t| > T/2 \end{cases}$$

The truncated signal is shown in Fig. 3.41. The integral on the right-hand side of Eq. (3.75) is the energy of the truncated signal $g_T(t)$. Thus,

$$P_g = \lim_{T \rightarrow \infty} \frac{E_{g_T}}{T} \quad (3.76)$$

- Signal power is time average of signal energy
- So, we can conclude that power spectral density PSD would be the time average of energy spectral density ESD

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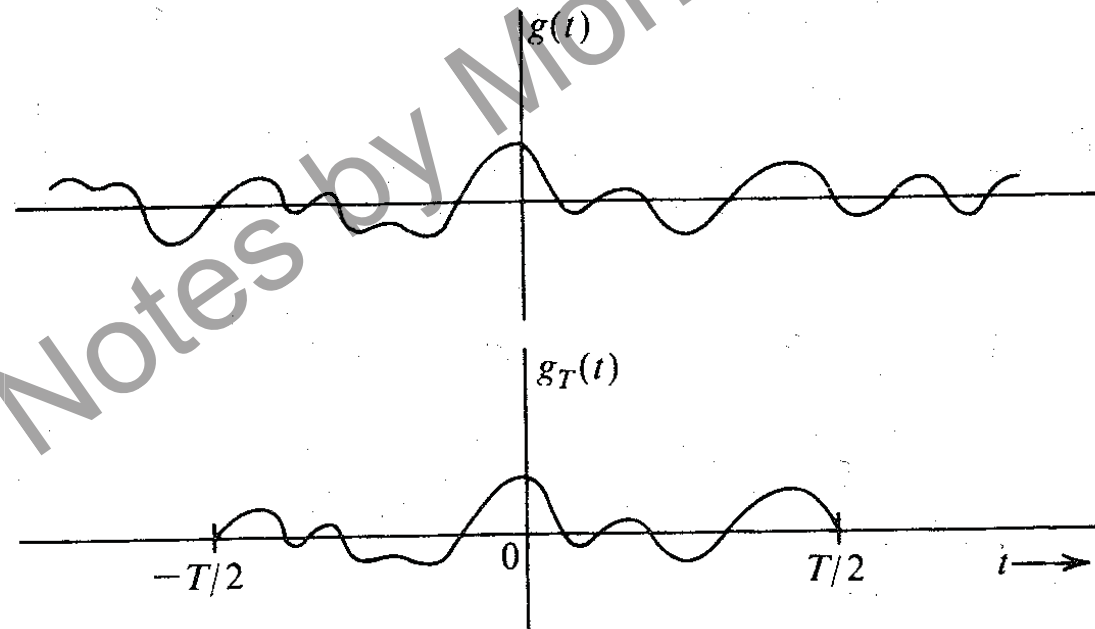


Figure 3.41 Limiting process in derivation of PSD.

Power Spectral Density (PSD) $S_g(\omega)$

$$P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g^2(t) dt$$

If the signal $g(t)$ is a power signal, then its power is finite, and the truncated signal $g_T(t)$ is an energy signal as long as T is finite. If $g_T(t) \longleftrightarrow G_T(\omega)$, then from Parseval's theorem,

$$E_{g_T} = \int_{-\infty}^{\infty} g_T^2(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |G_T(\omega)|^2 d\omega$$

Hence, P_g , the power of $g(t)$, is given by

$$P_g = \lim_{T \rightarrow \infty} \frac{E_{g_T}}{T} = \lim_{T \rightarrow \infty} \frac{1}{T} \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} |G_T(\omega)|^2 d\omega \right] \quad (3.77)$$

$$P_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} \lim_{T \rightarrow \infty} \frac{|G_T(\omega)|^2}{T} d\omega$$

We define the **power spectral density (PSD)** $S_g(\omega)$ as

$$S_g(\omega) = \lim_{T \rightarrow \infty} \frac{|G_T(\omega)|^2}{T}$$

Power Spectral Density (PSD) $S_g(\omega)$

$$P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g^2(t) dt$$

$$S_g(\omega) = \lim_{T \rightarrow \infty} \frac{|G_T(\omega)|^2}{T}$$

Consequently,*

$$P_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_g(\omega) d\omega \quad (3.80a)$$

$$= \frac{1}{\pi} \int_0^{\infty} S_g(\omega) d\omega \quad (3.80b)$$

This result is parallel to the result [Eq. (3.68)] for energy signals. The power is $1/2\pi$ times the area under the PSD. Observe that the PSD is the time average of the ESD of $g_T(t)$ [Eq. (3.79)].

- As is the case with ESD, the PSD is a positive, real and even function of ω .
- If $g(t)$ is a voltage signal, the units of PSD are volts squared per hertz.
- Using variable f in Hz, the above equations in more compact form;

$$P_g = \int_{-\infty}^{\infty} S_g(\omega) df = 2 \int_0^{\infty} S_g(\omega) df$$

Time Autocorrelation Function $\mathcal{R}_g(\tau)$ and Power Spectral Density (PSD) $S_g(\omega)$

The (time) autocorrelation function $\mathcal{R}_g(\tau)$ of a real power signal $g(t)$ is defined as[†]

$$\mathcal{R}_g(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g(t)g(t + \tau) dt \quad (3.82a)$$

Using the same argument as that used for energy signals [Eqs. (3.71b) and (3.72)], we can show that $\mathcal{R}_g(\tau)$ is an even function of τ . This means for a real $g(t)$

$$\mathcal{R}_g(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g(t)g(t - \tau) dt \quad (3.82b)$$

and

$$\mathcal{R}_g(\tau) = \mathcal{R}_g(-\tau) \quad (3.83)$$

$$\mathcal{R}_g(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\infty}^{\infty} g_T(t)g_T(t + \tau) dt = \lim_{T \rightarrow \infty} \frac{\psi_{g_T}(\tau)}{T}$$

Recall that $\psi_{g_T}(\tau) \iff |G_T(\omega)|^2$. Hence, the Fourier transform of the preceding equation yields

$$\mathcal{R}_g(\tau) \iff \lim_{T \rightarrow \infty} \frac{|G_T(\omega)|^2}{T} = S_g(\omega) \quad (3.84)$$

Interpretation of Power Spectral Density

Because the PSD is a time average of the ESD of $g(t)$, we can argue along the lines used in the interpretation of ESD. We can readily show that the PSD $S_g(\omega)$ represents the power per unit bandwidth (in hertz) of the spectral components at the frequency ω . The power contributed by the spectral components within the band ω_1 to ω_2 is given by

$$\Delta P_g = \frac{1}{\pi} \int_{\omega_1}^{\omega_2} S_g(\omega) d\omega \quad (3.87)$$

Table 3.3

$E_g = \int_{-\infty}^{\infty} g^2(t) dt$	$P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g^2(t) dt = \lim_{T \rightarrow \infty} \frac{E_{gT}}{T}$
$\psi_g(\tau) = \int_{-\infty}^{\infty} g(t)g(t+\tau) dt$	$\mathcal{R}_g(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g(t)g(t+\tau) dt = \lim_{T \rightarrow \infty} \frac{\psi_{gT}(\tau)}{T}$
$\Psi_g(\omega) = G(\omega) ^2$	$S_g(\omega) = \lim_{T \rightarrow \infty} \frac{ G_T(\omega) ^2}{T} = \lim_{T \rightarrow \infty} \frac{\Psi_{gT}(\omega)}{T}$
$\psi_g(\tau) \Longleftrightarrow \Psi_g(\omega)$	$\mathcal{R}_g(\tau) \Longleftrightarrow S_g(\omega)$
$E_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} \Psi_g(\omega) d\omega$	$P_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_g(\omega) d\omega$

Importance of Autocorrelation Method

- Signals in communication are **random** and can not be described as a function of time (indeterministic signals).
- Fourier transform is available only for deterministic signals and it generally does not exist for power signals.
 - For e.g. **a random binary pulse** train can not be described as a function of time, and it is impossible to find their Fourier transforms.
 - However, the **autocorrelation function** for such signals can be determined which also allows us to determine the **PSD** (*the spectral information*) of such a signal.

EXAMPLE 3.23 Figure 3.42a shows a random binary pulse train $g(t)$. The pulse width is $T_b/2$, and one binary digit is transmitted every T_b seconds. A binary 1 is transmitted by the positive pulse, and a binary 0 is transmitted by the negative pulse. The two symbols are equally likely and occur randomly. We shall determine the autocorrelation function, the PSD, and the essential bandwidth of this signal.

$$\mathcal{R}_g(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g(t)g(t - \tau) dt$$

Let there be N bits (pulses) during this interval T so that $T = N T_b$, and as $T \rightarrow \infty$, $N \rightarrow \infty$. Thus,

$$\mathcal{R}_g(\tau) = \lim_{N \rightarrow \infty} \frac{1}{N T_b} \int_{-N T_b/2}^{N T_b/2} g(t)g(t - \tau) dt$$

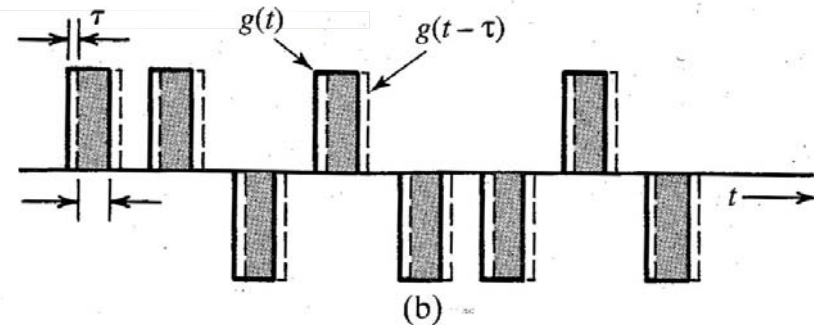
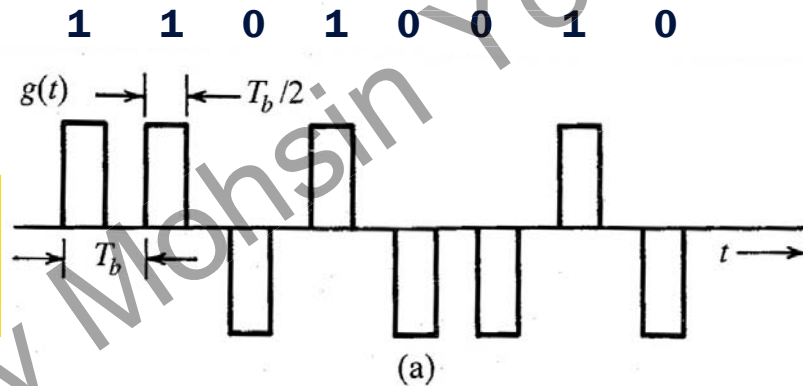
Case 1: $\tau < T_b/2$

The area under the product is:

$$g(t)g(t - \tau) = \frac{T_b}{2} - \tau$$

Since there are N pulses during the averaging interval, we have;

$$\mathcal{R}_g(\tau) = \lim_{N \rightarrow \infty} \frac{1}{N T_b} \left[N \left(\frac{T_b}{2} - \tau \right) \right]$$



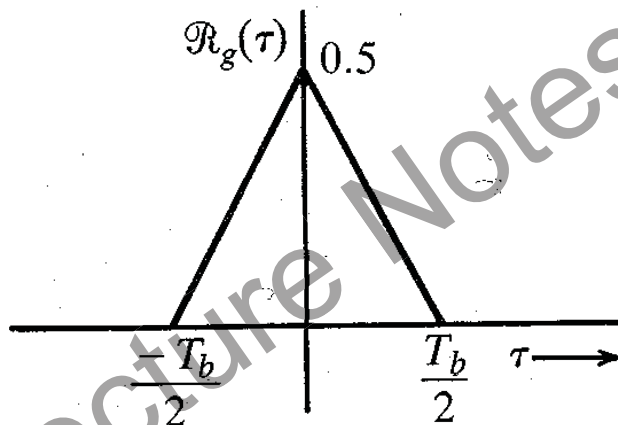
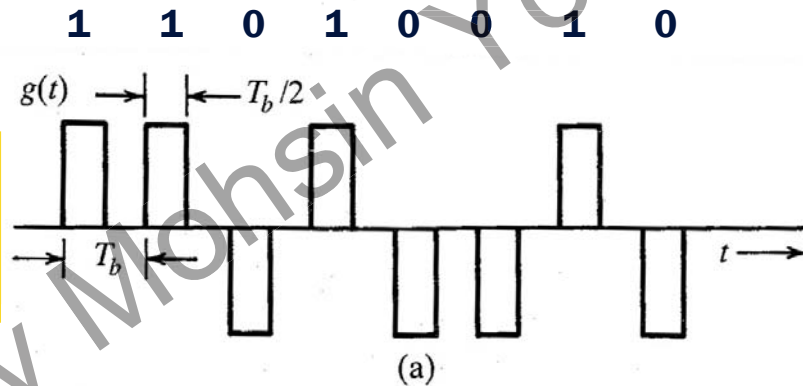
$$\mathcal{R}_g(\tau) = \frac{1}{2} \left(1 - \frac{2\tau}{T_b} \right), \quad \tau < \frac{T_b}{2}$$

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Case 1: $\tau < T_b/2$

Because $\mathcal{R}_g(\tau)$ is an even function of τ ,

$$\mathcal{R}_g(\tau) = \frac{1}{2} \left(1 - \frac{2|\tau|}{T_b} \right), \quad |\tau| < \frac{T_b}{2}$$



Case 2: $\tau > T_b/2$

As we increase τ beyond $T_b/2$, there will be overlap between each pulse and its immediate neighbor.

Consequently the area under $g(t)g(t - \tau)$ will be zero when averaged over an infinitely large time ($T \rightarrow \infty$), and

$$\mathcal{R}_g(\tau) = 0, \quad |\tau| > \frac{T_b}{2}$$

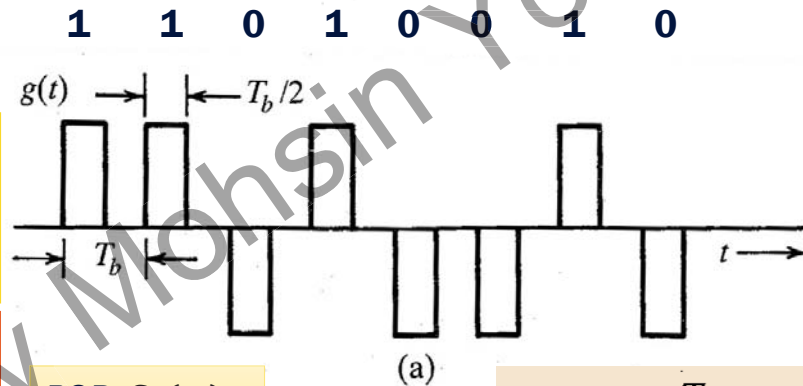
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Because $\mathcal{R}_g(\tau)$ is an even function of τ ,

$$\mathcal{R}_g(\tau) = \frac{1}{2} \left(1 - \frac{2|\tau|}{T_b} \right), \quad |\tau| < \frac{T_b}{2}$$

$$\mathcal{R}_g(\tau) = \frac{1}{2} \Delta \left(\frac{\tau}{T_b/2} \right), \quad |\tau| < \frac{T_b}{2}$$



PSD $S_g(\omega)$

$$S_g(\omega) = \frac{T_b}{4} \text{sinc}^2 \left(\frac{\omega T_b}{4} \right)$$

